HARE.jl

Heteroskedasticity and Autocorrelation Estimators – feasible GLS corrections for linear regression models.

HARE.jl provides classical estimators and tests covering heteroskedasticity, AR(1) autocorrelation, and their combination:

Error structure	Estimators / Tests
Exponential variance (Harvey)	`exponential_fwls`, `exponential_mle`, `harvey_test`
Power variance	`power_fwls`, `power_mle`
Quadratic variance (Glejser)	`quadratic_fwls`, `quadratic_mle`, `glejser_test`
Linear variance	`linear_fwls`, `linear_mle`
Groupwise heteroscedasticity	`two_step_groupwise`, `iterated_groupwise`, `groupwise_mle`
AR(1) autocorrelation	`two_step_prais_winsten`, `iterated_prais_winsten`, `two_step_cochrane_orcutt`, `iterated_cochrane_orcutt`, `hildreth_lu`
AR(1) + heteroskedasticity	`two_step_sequential`, `iterated_sequential`, `two_step_joint`, `iterated_joint`
AR(1) exact MLE	`beach_mackinnon`

All estimators return typed result structs that implement the StatsBase interface: `coef`, `stderror`, `vcov`, `residuals`, `fitted`, `predict`.

Installation

```julia
using Pkg
Pkg.add(url = "https://github.com/Trumpingtons/HARE.jl")
```

Quick Start

Pass regressors **without** a constant column – the intercept is added automatically:

```julia
using HARE, Random

Random.seed!(1234)
n   = 500
x1  = randn(n); x2 = randn(n)
X = hcat(x1, x2)
y   = hcat(ones(n), X) * [1.0, 2.0, -1.0] .+ exp.(0.5 .* x1) .* randn(n)

m = exponential_ifwls(X, y)
coef(m)      # coefficient vector
stderror(m)  # standard errors
vcov(m)      # full covariance matrix
residuals(m) # in-sample residuals
```

The `@formula` interface mirrors GLM.jl:

```julia
using HARE
data = (y = y, x1 = x1, x2 = x2)
m = exponential_ifwls(@formula(y ~ x1 + x2), data)
```

Contents

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Pages = ["tutorial.md", "api.md"]
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